Microsoft Elevate

CAPSTONE PROJECT

HR EMPLOYEE ATTRITION

PRESENTED BY

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PROBLEM STATEMENT:

Organizations face challenges in understanding employee attrition patterns due to large volumes of HR data stored in raw tabular formats. Traditional data analysis methods make it difficult for HR managers to quickly identify key factors such as department-wise attrition, salary trends, age distribution, and job satisfaction levels. The absence of interactive and visual analytical tools limits timely decision-making, leading to higher employee turnover, increased recruitment costs, and reduced organizational productivity. Therefore, there is a need for an effective system that can analyze HR data and present meaningful insights in an intuitive manner.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of analyzing and understanding employee attrition within an organization to support effective human resource decision-making. The system focuses on transforming raw HR data into meaningful insights through interactive data analytics and visualization techniques. By identifying patterns related to employee demographics, job roles, salary structure, and job satisfaction, the system helps organizations understand the key factors influencing employee turnover.
The solution consists of the following components:

1. Data Collection
Historical employee data is collected in the form of a structured dataset containing details such as employee age, gender, department, job role, monthly income, years of experience, job satisfaction, performance rating, and attrition status. The dataset serves as the primary source of information for analyzing workforce behavior and attrition trends.

2. Data Preprocessing
The collected data is cleaned and preprocessed using data transformation techniques. This includes handling missing values, removing duplicate records, correcting data types, and standardizing categorical fields such as department, gender, and attrition status. Data preprocessing ensures accuracy, consistency, and reliability of the analysis.

3. Data Analytics & Measures
Key analytical measures are created using Data Analysis Expressions (DAX) to compute important HR metrics such as total employees, attrition count, attrition rate, average age, average monthly income, and average years of experience. These measures enable dynamic and context-aware analysis based on user-selected filters.

4. Visualization & Dashboard Design
An interactive dashboard is designed using Power BI to visually represent workforce trends and attrition patterns. Various charts such as bar charts, pie charts, scatter plots, and KPI cards are used to display department-wise employee distribution, age vs income analysis, gender distribution, and attrition trends. Slicers are added to allow users to filter data dynamically by department, gender, job role, and age group.

PROPOSED SOLUTION:

5. Deployment
The completed HR Analytics Dashboard can be deployed using Microsoft Power BI Service, enabling secure access through a Microsoft account. The system supports scalability and can be integrated with enterprise databases such as Azure SQL for real-time HR analytics and reporting.

6. Evaluation
The effectiveness of the system is evaluated based on the accuracy and clarity of insights generated from the dashboard. User interaction with slicers and visuals is monitored to ensure correct filtering and meaningful analysis. Feedback from users can be used to enhance dashboard design, add new KPIs, and improve usability.

7. Result
The final system provides a comprehensive HR Analytics Dashboard that offers clear insights into employee attrition trends, workforce demographics, and salary distribution. The dashboard enables HR managers to identify high-risk attrition areas, understand employee behavior, and support data-driven workforce planning decisions.

SYSTEM APPROACH:

System Approach
The System Approach section describes the overall methodology adopted to design, develop, and implement the HR Analytics Employee Attrition Dashboard. The approach focuses on transforming raw HR data into meaningful insights using data analytics, visualization techniques, and interactive reporting tools. The system is designed to support human resource professionals in analyzing workforce trends and identifying attrition patterns efficiently.

1. System Requirements
Hardware Requirements
- Personal Computer or Laptop
- Minimum 4 GB RAM (8 GB recommended)
- Intel i3 processor or higher
- At least 5 GB free disk space

Software Requirements
- Operating System: Windows 10 or higher
- Microsoft Power BI Desktop
- Microsoft Excel / CSV file support
- Web browser (for Power BI Service access)

2. Libraries / Tools Required to Build the System
Since the project is developed using Power BI, the following tools and components are used:
- **Power BI Desktop** – Used for data loading, transformation, modeling, and dashboard creation
- **Power Query Editor** – Used for data cleaning, preprocessing, and transformation
- **DAX (Data Analysis Expressions)** – Used to create calculated measures such as attrition rate, average salary, and employee count
- **Power BI Service** (Optional) – Used for dashboard deployment and sharing
- **CSV Dataset** – Used as the data source for HR employee records

ALGORITHM & DEPLOYMENT:

Algorithm
This section describes the analytical approach used to process HR data and derive meaningful insights related to employee attrition. Instead of a traditional machine learning prediction model, the system uses a structured data analytics algorithm implemented through Power BI and DAX to compute key performance indicators and analyze attrition patterns dynamically.

1. Algorithm Selection
The project adopts a **descriptive and diagnostic analytics approach** using **DAX (Data Analysis Expressions)** within Microsoft Power BI. This approach is selected because the problem focuses on **analyzing historical employee data**, identifying attrition trends, and supporting decision-making rather than forecasting future values. DAX is well-suited for this purpose as it enables context-aware calculations, dynamic filtering, and real-time aggregation of HR metrics based on user interactions.

2. Data Input
The algorithm uses structured HR employee data as input. The key input features include:
- Employee ID
- Age and Age Group
- Gender
- Department
- Job Role
- Monthly Income
- Years at Company
- Total Working Years
- Job Satisfaction
- Performance Rating
- Overtime Status
- Attrition Status (Yes/No)

These features are used to analyze workforce distribution, salary patterns, and factors influencing employee attrition.

ALGORITHM & DEPLOYMENT:

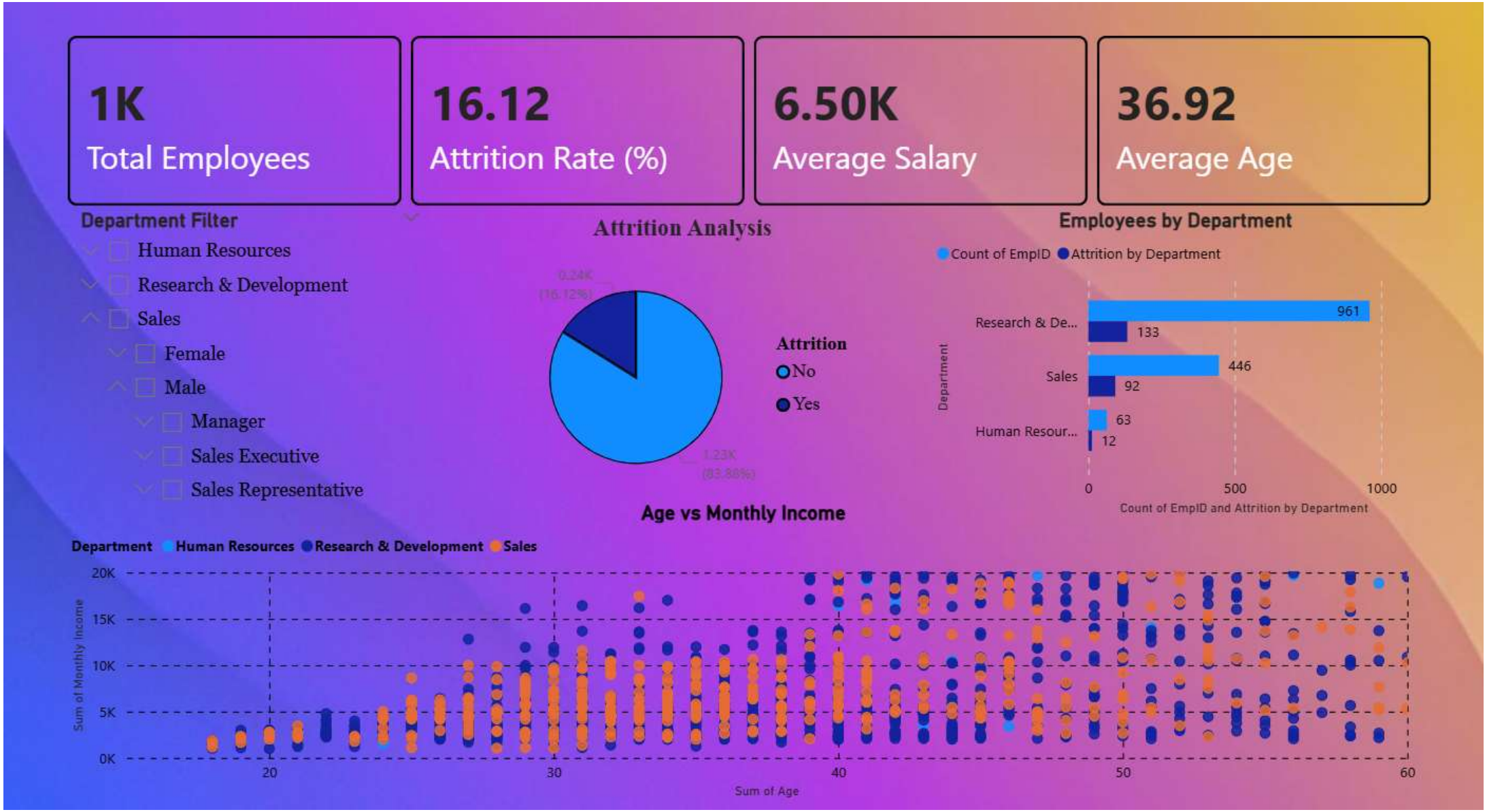
3. Training / Computation Process
The system does not require traditional model training. Instead, it applies **predefined analytical rules and DAX measures** to compute metrics from historical data. The process includes:
Cleaning and transforming data using Power Query
Defining calculated measures such as total employees, attrition count, and attrition rate using DAX
Applying aggregation and filtering logic based on selected dimensions (department, gender, job role, age group)
Validating calculated outputs through interactive filtering
This approach ensures accurate and consistent computation across all visuals.

4. Prediction / Analysis Process
The analysis process is performed dynamically at runtime. When users interact with slicers or filters, the DAX measures are recalculated in real time based on the selected context. This allows HR managers to analyze attrition trends across different employee segments instantly. The system provides immediate insights into high-risk attrition areas without requiring manual recalculation or external processing.

RESULT:

- The results of the HR Analytics Employee Attrition project demonstrate the effectiveness of the system in analyzing workforce data and identifying key attrition patterns. The interactive dashboard successfully converts raw HR data into meaningful visual insights using calculated KPIs and dynamic filtering.
- **Key Outcomes**
- The dashboard accurately displays the **total number of employees**, **attrition count**, and **attrition rate**, allowing HR managers to quickly assess overall workforce stability.
- Department-wise analysis reveals clear differences in employee distribution and attrition trends across departments.
- Visual comparison between employee age and monthly income highlights salary variations across different age groups.
- Gender-based and job-role-based filtering enables detailed demographic analysis.
- Attrition trends are clearly observable for employees with lower job satisfaction and fewer years at the company.
- **Visualization-Based Validation**
- The effectiveness of the system is validated through visual comparison and interaction:
- KPI cards dynamically update based on slicer selections, ensuring correct aggregation.
- Bar and column charts clearly represent employee counts and attrition distribution.
- Scatter plots visually demonstrate relationships between age and income.
- Slicers allow real-time filtering, confirming the correctness and responsiveness of the analytical measures.

RESULT:


1K
Total Employees
16.12
Attrition Rate (%)
6.50K
Average Salary
36.92
Average Age
Department Filter
Human Resources
Research & Development
Sales
Female
Male
Manager
Sales Executive
Sales Representative
Attrition Analysis
Attrition
No
Yes
Employees by Department
Count of EmpID
Attrition by Department
Research & De...
961
133
Sales
446
92
Human Resour...
63
12
Department
0
500
1000
Count of EmpID and Attrition by Department
Age vs Monthly Income
Department
Human Resources
Research & Development
Sales
20K
15K
10K
5K
0K
Sum of Monthly Income
20
30
40
50
60
Sum of Age

CONCLUSION:

The analysis revealed noticeable variations in attrition across different departments and job roles. Employees with lower job satisfaction levels and fewer years of experience showed higher attrition rates. Salary and age analysis highlighted disparities that may influence employee retention. The use of interactive slicers enabled detailed exploration of data based on department, gender, job role, and age group, making the insights more actionable.

FUTURE SCOPE:

- The HR Analytics Employee Attrition Dashboard can be further enhanced in several ways to increase its effectiveness and applicability in real-world organizational environments. Future improvements may include:
- Integration with **real-time HR databases** such as Azure SQL or organizational HRMS systems to enable live data updates.
- Implementation of **machine learning models** to predict employee attrition risk based on historical trends and employee behavior.
- Development of **role-based dashboards** for HR managers, department heads, and executives to provide customized insights.
- Inclusion of additional KPIs such as employee engagement scores, training effectiveness, and promotion trends.
- Automated alert systems to notify HR teams about departments or roles with unusually high attrition.
- Deployment of the dashboard across cloud platforms to support scalability and enterprise-wide access.

REFERENCES:

- Microsoft Power BI Documentation
- Kaggle – HR Analytics Employee Attrition Dataset
- Microsoft Learn – Data Analysis Expressions (DAX) Guide
- HR Analytics and Workforce Management Concepts
- Data Visualization Best Practices – Microsoft

GitHub Link: https://github.com/scotchash/HR-ANALYSIS-DASHBOARD-POWERBI-2

Thank You